Multi Variable Calculus.

Single no output

$f(x) = x^2$

Single variable input

a) $f(x,y) = x^2 + y^2$ b) $f(x,y) = (y,x)$

both are multivariable fns.

for a) for each (x & y) we get a number

b) There is a vector eg $(1,2) - (1i + 2j)$

$f(x,y) = x^2 + y$

$f(x,y) = \begin{bmatrix} 3x \\ 2y \end{bmatrix}$

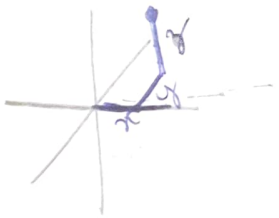
multiple variables as input.

vector

$f(x,y) = x^2 + y^2 = z$ ↳ to a number.

$f(1,1) = 2$

$f(2,2) = 8$ } →



$(x, y, z = x^2 + y^2)$ ⇒ graph 3D.



$(x, y, z = x + y)$ ⇒ graph 2D ⇒ plane ⇒ paper.

Curve

* has length only

Surface

*

If x and y are given as functions

$x = f(t)$

$y = g(t)$ over an interval of 't' values. Then, $(x,y) = (f(t), g(t))$ is called a parametric curve.

The eqns associated are called parametric equations.

Q: Sketch the curve defined by the parametric equations

$f(t) = 1 + 2t$

$g(t) = -3t$

Ans:

t	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
$f(t)$	-1	0	1	2	3
$g(t)$	3	$\frac{3}{2}$	0	$-\frac{3}{2}$	-3

$1+(2\times-1) = 1-2 = -1$

$1+(2\times-\frac{1}{2}) = 1-1 = 0$

$1+0 = 1$

$1+1 = 2$

$1+(2\times1) = 3$

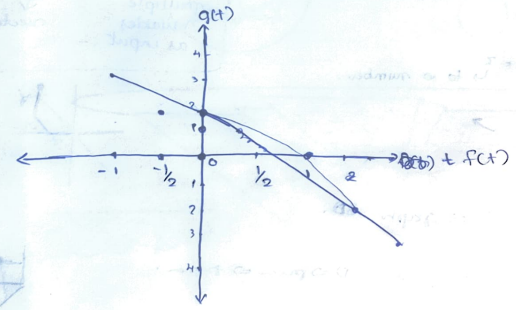
$(-1, 3)$

$(0, 1.5)$

$(1, 0)$

$(2, -1.5)$

$(3, -3)$





Q: $f(t) = t^2$

$g(t) = 2t - 1$

t	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
$f(t)$	1	$\frac{1}{4}$	0	$\frac{1}{4}$	1
$g(t)$	-3	-2	-1	0	1

$(1, 1)$

$(\frac{1}{4}, -2)$

$(0, -1)$

$(\frac{1}{4}, 0)$

$(1, 1)$

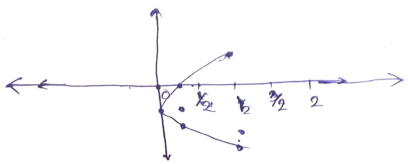
$-2-1$

$-2\times-\frac{1}{2}$

$1-1$

$2\times\frac{1}{2}$

$-1-1$



Q: Find the parametric equation for a line (a, b) and having a slope 'm'.

cartesian eqn: $y - b = m(x - a)$

if $x = t = f(t)$

$y - b = m(t - a)$

$y = b + m(t - a)$

$\therefore f(t) = t$

$g(t) = b + m(t - a)$

$-\infty < t < \infty$

why $f(t)$ is put 't'?

we can also use $f(t) = t$

$f(t) = t^3$

$g(t) = b + m(t^3 - a)$

$-\infty < t < \infty$

for same t, we get diff points. But still we get the same unique line at last.

Q: write the parametric representation of the line segment with end points $(0,0)$ and $(1,1)$.

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 0}{x - 0} = \frac{1 - 0}{1 - 0}$$

$$\frac{y}{x} = 1$$

$y = x$ cartesian

① $x = t = f(t)$ $\quad 0 \le t \le 1$

$y = g(t) = t$

② $x = t^2 = f(t)$ $\quad 0 \le t \le 1$

$y = g(t) = t^2$

$x = t = f(t)$

$y = x$

$g(t) = f(t)$

$y - x = 0$

$f(t) - f(t) = 0$

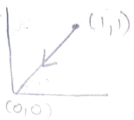
$\frac{y-1}{x-1} = \frac{-1}{-1}$ (1,1) (0,0)

$y-1 = x-1$

$y - x = -1+1$

$y - x = 0$

$y = x$



× $x = t = f(t)$
$y = g(t) = t$ $0 \le t \le 1$

here we get for $f(t) = 0$, $g(t) = 0$
tho we should be getting $f(t) = 1$ & $g(t) = 1$

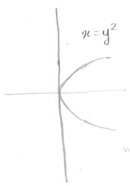
✓ $f(t) = x = 1 - t$
$g(t) = y = 1 - t$ $x = y$ $0 \le t \le 1$

$f(t) = x = -t$
$g(t) = y = -t$ $x = y$ $-1 \le t \le 0$

J

Why parametric eqn.?

Q: parametric eqn of a curve $x = y^2$?



$x = f(t) = t^2$
$y = g(t) = t$

$-\infty < t < \infty$

* when $-1 \le t \le 1$

$x = f(t) = t^2$
$y = g(t) = t$

to change the orientation



* $x = f(t) = (1-t)^2$
$y = g(t) = (1-t)$

$0 \le t \le 2$.

* $x = t^2$ $y = -t$

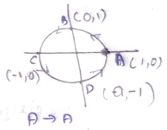
$-1 \le t \le 1$

diff parametrics for same parabola.

$\frac{1-t}{(1-t)^2}$

$x^2 + y^2 = 1$
$(x, f(x))$

$x = t$ $y = \pm\sqrt{1-t^2}$ not a fn



$P \to A$

$\sin^2 t + \cos^2 t = 1$

$x = \sin t$ $y = \cos t$

for any t,

B// $\sin t = 0$
$\cos t = 1$

A// $x = \sin t$ $y = \cos t$ $\sin t = 1$ $\cos t = 0$

$\pi/2 \le t \le 5\pi/2$ $5\pi/2 - 2\pi = \pi/2$

$2\pi \le t$

not good parametrization

C// $\sin t = -1$
$\cos t = 0$

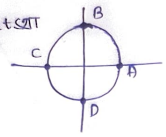
$x = \cos t$ $y = \sin t$
$0 \le t \le 2\pi$

D// $\sin t = 0$
$\cos t = -1$

Q: Give a parametrisation of the circle $x^2+y^2=1$

HW

Starting from



$0 \le t \le 2\pi$

$x = a\cos t$
$y = -a\sin t$ 1) ADCBA

2) BADCB $x = a\sin t$, $y = a\cos t$

$x = -a\sin t$
$y = -a\cos t$ 3) DCBAD

4) ABCDA → $x = a\cos t$, $y = a\sin t$, $0 \le t \le 2\pi$

circle

$x^2+y^2=a^2$

$x = a\cos t$, $y = a\sin t$

$0 \le t \le 2\pi$

ellipse.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Q: Find parametric eqn eq the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Q: hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(refer hyperbolic trigonometric fns)

parametrics of ellipse:

$0 \le t \le 2\pi$

$x = a\cos t$

$y = b\sin t$

? $x(t) = \cos t$

$y(t) = \sin t$ $\quad 0 \le t \le 2\pi$

$z(t) = t$

Shape: Circular coil Spring

* Suppose $x = f(t)$ $y = g(t)$ $z = h(t)$ $t \in I$. Then, the same curve represented by this parametric eqn can be represented in a vector form.

ie $r(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$

Let $r(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$ be a vector function in the domain D and let L be a vector. r has a limit L as t approaches t_0, $\lim_{t \to t_0} r(t) = \vec{L}$ ⇒ limit of vectors will be a vector

Set of vectors tending to a vector.

* A fn $r(t)$ is continuous at $t = t_0$ if $\lim_{t \to t_0} r(t) = r(t_0)$

* A vector fn r is said to be differentiable, if f, g & h are differentiable at t and the derivative is given by $r'(t) = \frac{dr}{dt} = \frac{df}{dt}\hat{i} + \frac{dg}{dt}\hat{j} + \frac{dh}{dt}\hat{k}$

★ Lines & planes in space 732 (12.5)

★ Cylinders & Quadric Surfaces (740) (12.6)

Surface

$x^2+y^2+z^2 = 1$

If $r(t) = (\cos t)\hat{i} + (\sin t)\hat{j} + tk$, then

$$\lim_{t \to \pi/4} r(t) = \left(\lim_{t \to \pi/4} \cos t\right)\hat{i} + \left(\lim_{t \to \pi/4} \sin t\right)\hat{j} + \left(\lim_{t \to \pi/4} t\right)k$$

$$= \frac{\sqrt{2}}{2}\hat{i} + \frac{\sqrt{2}}{2}\hat{j} + \frac{\pi}{4}\hat{k}$$

$$= \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{\pi}{4}\hat{k}.$$

Ex: $r(t) = 2\cos t\,\hat{i} + 2\sin t\,\hat{j} + 5\cos^2 t\,\hat{k}$

Vel, speed, acc

$v(-\pi/4)$

$$\lim_{t \to -\pi/4} r(t) = \left(\lim_{t \to -\pi/4} 2\cos t\right)\hat{i} + \left(\lim_{t \to -\pi/4} 2\sin t\right)\hat{j} + \left(\lim_{t \to -\pi/4} 5\cos^2 t\right)\hat{k}$$

$$v(t) = \frac{d}{dt}(2\cos t)\hat{i} + \frac{d}{dt}(2\sin t)\hat{j} + \frac{d}{dt}(5\cos^2 t)\hat{k}$$

$$= 2\sin t\,\hat{i} + 2\cos t\,\hat{j} + 5 \times 2 \times \cos t \times -\sin t\,\hat{k}$$

$$= 2\sin t\,\hat{i} + 2\cos t\,\hat{j} + 10\cos t \cdot \sin t\,\hat{k}$$

$$v(-\pi/4) = 2\sin(-\pi/4)\hat{i} + 2\cos(-\pi/4)\hat{j} + 10\cos(-\pi/4)\cdot\sin(-\pi/4)\hat{k}$$

$$= -\frac{2}{\sqrt{2}}\hat{i} + \frac{2}{\sqrt{2}}\hat{j} - 10 \times \cos$$

$$a(t) = 2\cos t\,\hat{i} + -2\sin t\,\hat{j} - 10$$

Q: let $r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$

Find $\lim_{t \to 1}\left[r(t) \cdot (r'(t)) \times (r''(t))\right]$

$$r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$$

$$r'(t) = 1\hat{i} + 2t\hat{j} + 3t^2\hat{k}$$

$$r''(t) = 2\hat{j} + 6t\hat{k}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix}$$

$$r'(t) \times r''(t) = \hat{i}(12t^2 - 6t^2) - \hat{j}(6t - 0) + 2\hat{k}$$

$$= (12t^2 - 6t^2)\hat{i} - \hat{j}(6t - 3t^2) + \hat{k}(2 - 0)$$

$$= 6t^2\hat{i} - 6t\hat{j} + 2\hat{k}$$

$$= 6t^2\hat{i} - 6t\hat{j} + 2\hat{k}$$

$$r(t) \cdot (r'(t) \times r''(t)) = (2\hat{j} + 6t\hat{k}) \cdot (6t^2\hat{i} - 6t\hat{j} + 2\hat{k})$$

$$= 0\hat{i} + 0\hat{j} + 12t\hat{k}$$

$$\lim_{t \to 1}\left[r(t) \cdot (r'(t) \times r''(t))\right] = 12 \times 1\,\hat{k} = 12\hat{k}$$

Q: $\lim_{t \to 1} r(t) \cdot \lim_{t \to 1}\left(r'(t) \times r''(t)\right)$

$$= (1\hat{i} + 1\hat{j} + 1\hat{k}) \cdot (6\hat{i} - 6\hat{j} + 2\hat{k})$$

$$= 6\hat{i} + 6\hat{j} + 2\hat{k}$$

$$= 6 - 6 + 2 = 2$$

① $\lim_{t\to 1}\left[r(t)\cdot(r'(t)\times r''(t))\right]$

②

$$r(t)\cdot(r'(t)\times r''(t)) = (t\hat{i}+t^2\hat{j}+t^3\hat{k})\cdot(6t^2\hat{i}-6t\hat{j}+2\hat{k})$$

$$= 6t^3 - 6t^3 + 2t^3$$

$$\lim_{t\to 1} \quad 6t^3+6t^3+2t^3 = 6-6+2 = \underline{2}$$

③ $\lim_{t\to 1} r(t)\cdot\left(\lim_{t\to 1} r'(t) \times \lim_{t\to 1} r''(t)\right)$

$$\lim_{t\to 1} r(t) = 1+1+1 = \underline{3}$$

$$\lim_{t\to 1} r'(t) = \lim_{t\to 1} 1\hat{i}+2t\hat{j}+3t^2\hat{k}$$

$$= 1\hat{i}+2\hat{j}+3\hat{k} = \underline{6}$$

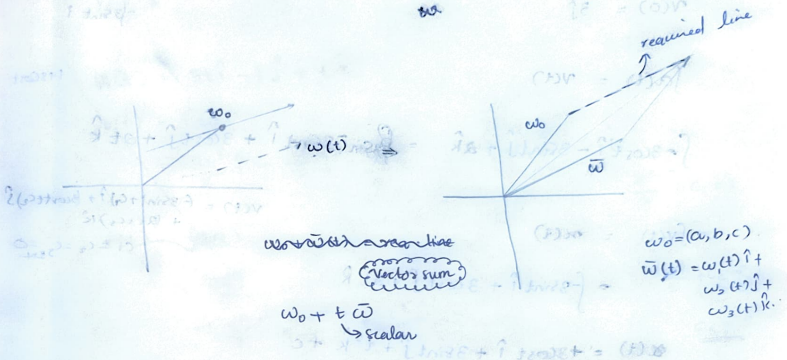
$$\lim_{t\to 1} r''(t) = \lim_{t\to 1} 2\hat{j}+6t\hat{k}$$

$$= 0+2\hat{j}+6\hat{k}$$

$$= \underline{8}$$

$\lim_{t\to 0} r'(t) \times \lim_{t\to 1} r''(t) = (1\hat{i}+2\hat{j}+3\hat{k})\times(0\hat{i}+2\hat{j}+6\hat{k})$

1 2 3
0 2 6



Q: Find the parametric eqn for the line that is tangent to the given curve.

$r(t) = t^2\hat{i} + (2t-1)\hat{j} + t^3\hat{k}$ at $t_0 = 2$.

$\frac{d(2t-1)}{dt} = 2$

$r(t_0) = 4\hat{i} + 3\hat{j} + 8\hat{k} \Rightarrow \omega_0$

$r'(t) = 2t\hat{i} + 2\hat{j} + 3t^2\hat{k} \Rightarrow \omega$

~~$\omega_0 + t\omega = (4\hat{i}+3\hat{j}+8\hat{k}) + t(2t\hat{i}+2\hat{j}+3t^2\hat{k})$~~

~~$= (2t^2+4)\hat{i} + (2t+3)\hat{j} + (3t^3+8)\hat{k}$~~

$r'(2) = 0\hat{i}+0\hat{j}+0\hat{k} = 4\hat{i}+2\hat{j}+12\hat{k} \Rightarrow \omega$

$a(t) = -3\cos t\,\hat{i} - 3\sin t\,\hat{j} + 2\hat{k}$.

at $t=0$, $x(0) = (4,0,0)$

$v(0) = 3\hat{j}$

$\int a(t) = v(t)$

$\int -3\cos t\,\hat{i} - 3\sin t\,\hat{j} + 2\hat{k} = -3\sin t\,\hat{i} + 3\cos t\,\hat{j} + 2t\,\hat{k}$

$\int v(t) = x(t)$

$= \int -3\sin t\,\hat{i} + 3\cos t\,\hat{j} + 2t\,\hat{k}$

$x(t) = +3\cos t\,\hat{i} + 3\sin t\,\hat{j} + t^2\hat{k} + c$

$r(t) = (3\cos t + c_1)\,\hat{i} + (3\sin t + c_2)\hat{j} + (t^2 + c_3)\hat{k}$

at $t=0$

$3\cos t + c_1 = 4$ $\quad c_2 = 0$

$3\times\cos 0 + c_1 = 4$ $\quad c_3 = 0$

$c_1 = 1$

$r(t) = (1+3\cos t)\,\hat{i} + 3\sin t\,\hat{j} + t^2\hat{k}$

Q: At time $t=0$, a particle is located at $(1,2,3)$. It travels in a straight line to $(4,1,4)$ and has a speed of 2 initially and a constant acceleration $3\hat{i} - \hat{j} + \hat{k}$. Find the equation of the position vector?

$a = 3\hat{i} - \hat{j} + \hat{k}$

$v_0 = 2$

$v(t) = \int a(t)$

$= \frac{3t^2}{2}\hat{i} - \frac{t^2}{2}\hat{j} + \frac{t^2}{2}\hat{k} + c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$r(t) = \int v(t)$

$= \frac{t^3}{2}\hat{i} - \frac{t^3}{6}\hat{j} + \frac{t^3}{6}\hat{k}$

$= \left(\frac{t^3}{2} + k_1\right)\hat{i} - \left(\frac{t^3}{6} + k_2\right) + \left(\frac{t^3}{6} + k_3\right)\hat{k}$

$= \left(\frac{t^3}{2} + 1\right)\hat{i} - \left(\frac{t^3}{6} + 2\right)\hat{j} + \left(\frac{t^3}{6} + 3\right)\hat{k}$

$v(t) = \left(\frac{3t^2}{2} + c_1\right)\hat{i} - \left(\frac{t^2}{2} + c_2\right)\hat{j} + \left(\frac{t^2}{2} + c_3\right)\hat{k}$

$(1,2,3) \longrightarrow (4,1,4)$

$\Delta\vec{r} = (3,-1,1) = 3\hat{i} - 1\hat{j} + 1\hat{k}$

$|\vec{r}| = \sqrt{11}$

$\frac{t^3}{2} + c_1 = 1$ $\quad c_1 = 1$

$\frac{t^3}{6} + c_2 = 2$ $\quad c_3 = 3$

$$\frac{\vec{dr}}{|\vec{dr}|} = \frac{3}{\sqrt{11}}\hat{i} - \frac{1}{\sqrt{11}}\hat{j} + \frac{1}{\sqrt{11}}\hat{k}$$

$$\vec{v} = m\frac{\vec{r}}{|r|}$$

$$= 2\left(\frac{3}{\sqrt{11}}\hat{i} - \frac{1}{\sqrt{11}}\hat{j} + \frac{1}{\sqrt{11}}\hat{k}\right)$$

$$\vec{v} = \frac{6}{\sqrt{11}}\hat{i} - \frac{2}{\sqrt{11}}\hat{j} + \frac{2}{\sqrt{11}}\hat{k}$$

$$\therefore \frac{3t^2}{2} + c_1 = \frac{6}{\sqrt{11}} \qquad c_2 = \frac{2}{\sqrt{11}} \qquad c_3 = \frac{2}{\sqrt{11}}$$

$$c_1 = \frac{6}{\sqrt{11}}$$

$$\vec{v}(t) = \left(\frac{3t^2}{2} + \frac{6}{\sqrt{11}}\right)\hat{i} - \left(\frac{t^2}{2} + \frac{2}{\sqrt{11}}\right)\hat{j} + \left(\frac{t^2}{2} + \frac{2}{\sqrt{11}}\right)\hat{k}$$

$$\vec{r}(t) = \int v(t)$$

$$= \left(\frac{3t^3}{6} + \frac{6}{\sqrt{11}}t\right)\hat{i} - \left(\frac{t^3}{6} + \frac{2}{\sqrt{11}}t\right)\hat{j} + \left(\frac{t^3}{6} + \frac{2}{\sqrt{11}}t\right)\hat{k}$$

$r(\vec{t}) =$

$$a = 3\hat{i} - 1\hat{j} + 1\hat{k}$$

$$v(t) = 3t\hat{i} - t\hat{j} + t\hat{k} + c$$

$$v(t) = (3t + c_1)\hat{i} - (t + c_2)\hat{j} +$$

Q: A particle has acceleration $a(t) = -\sin t\,\hat{i} + \cos t\,\hat{j} + 3\hat{k}$. Its initial velocity $v(0)$ is given as $v(0) = \hat{j}$ and initial position $r(0)$ is given as $r(0) = \hat{i} + 2\hat{k}$. Find the eqn of position vector.

$\cos t\,\hat{j} \Rightarrow -\sin t$

$$v(t) = \int a(t)$$

$$= (+\cos t + c_1)\hat{i} + (+\sin t + c_2)\hat{j} + (3t + c_3)\hat{k}$$

$$v(0) = \hat{j} \qquad 0\hat{i} + 1\hat{j} + 0\hat{k}$$

$$\therefore v(0) = (+\cos 0 + c_1)\hat{i} + (+\sin 0 + c_2)\hat{j} + (3\times 0 + c_3)\hat{k}$$

$$= +1 + c_1\,\hat{i} + (0 + c_2)\,\hat{j} + (0 + c_3)\,\hat{k}$$

$\Downarrow 0\hat{i}$ $\Downarrow 1\hat{j}$ $\Downarrow 0\hat{k}$

$\therefore c_1 = -1 \qquad c_2 = 1 \qquad c_3 = 0$

$$\text{a } r(t) = \int v(t)$$

$$= \int (+\cos t - 1)\hat{i} + (+\sin t + 1)\hat{j} + (3t)\hat{k}$$

$$r(t) = (\sin t - t)\hat{i} + (\cos t + t)\hat{j} + \frac{3t^2}{2}\hat{k}$$

$$r(t) = (\sin t - t + k_1)\hat{i} + (\cos t + t + k_2)\hat{j} + \left(\frac{3t^2}{2} + k_3\right)\hat{k}$$

~~$\sin t - t + k$~~

$$r(0) = (0 - 0 + k_1)\hat{i} + (1 + 0 + k_2)\hat{j} + (0 + k_3)\hat{k}$$

$r(0) = \hat{i} + 0\hat{j} + 2\hat{k}$

$k_1 = 1 \qquad k_2 + 1 = 0 \qquad k_3 = 2$

$k_2 = +1$

$$r(t) = (\sin t - t + 1)\hat{i} + (-\cos t + t + 1) + \left(\frac{3t^2}{2} + 2\right)\hat{k}$$

Q: Find the length of the circle $x^2 + y^2 = 1$ from $(1,0)$ to (

parametrisation',

$f(t) = \cos t$

$g(t) = \sin t$

$0 \le t \le \pi$

$$L = \int_0^{\pi} \sqrt{(\sin t)^2 + (\cos t)^2}\, dt$$

$$= \int_0^{\pi} dt$$

$$= t\Big]_0^{\pi}$$

$$= \pi$$

The length of a smooth curve;

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

Q:- Find the length of the glider from $t=0$ to $t=2\pi$ with a position vector $r(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\,\hat{k}$

$$L = \int_0^{2\pi} \sqrt{\left(\frac{d(\cos t)}{dt}\right)^2 + \left(\frac{d(\sin t)}{dt}\right)^2 + \left(\frac{d(t)}{dt}\right)^2}\, dt \qquad \frac{d}{dt}(t) = 1$$

$$= \int_0^{2\pi} \sqrt{(\sin t)^2 + (\cos t)^2 + 1}\, dt$$

$$= \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t + 1}\, dt$$

$$= \int_0^{2\pi} \sqrt{1+1}\, dt$$

$$= \sqrt{2} \times t \Big]_0^{2\pi}$$

$$= \sqrt{2} \times 2\pi$$

$$= \underline{2\sqrt{2}\,\pi}$$

Q:- Find the unit tangent vector of the chorel;

$r(t) = (1+3\cos t)\hat{i} + (3\sin t)\hat{j} + t^2\hat{k}$

$r(t) = (1+3\cos t)\hat{i} + (3\sin t)\hat{j} + t^2\hat{k}$

$V(t) = (0+-3\sin t)\hat{i} + 3\cos t\,\hat{j} + 2t\,\hat{k}$ or

$V(t) = -3\sin t\,\hat{i} + 3\cos t\,\hat{j} + 2t\,\hat{k}$

unit vector $= \dfrac{\vec{v}}{|\vec{v}|}$

$$|\vec{v}| = \sqrt{(-3\sin t)^2 + (3\cos t)^2 + (2t)^2}$$

$$= \sqrt{9\sin^2 t + 9\cos^2 t + 4t^2}$$

$$= \sqrt{9(\sin^2 t + \cos^2 t) + 4t^2}$$

$$= \sqrt{9 + 4t^2}$$

unit vector; $\dfrac{-3\sin t\,\hat{i} + 3\cos t\,\hat{j} + 2t\,\hat{k}}{\sqrt{9+4t^2}}$

T is unit vector of a smooth curve.

curvature, $\kappa = \left|\dfrac{dT}{ds}\right|$

$$\kappa = \left|\frac{dT}{ds}\right| = \left|\frac{dT}{dt} \cdot \frac{dt}{ds}\right|$$

$$= \frac{1}{|ds/dt|} \left|\frac{dT}{dt}\right|$$

$$\kappa = \frac{1}{|V|} \frac{dT}{dt}$$

or

$T = \dfrac{V}{|V|}$ is the unit tangent vector."

Q: Find ... (1,1,0) & (2,1,3)

$\Delta r = (2-1)\hat{i} + (1-1)\hat{j} + (3-0)\hat{k}$

$= 1\hat{i} + 0\hat{j} + 3\hat{k}$

$\bar{r}(t) = (1+1t)\hat{i} + 1\hat{j} - 3t\hat{k}$

$x = 1+1t$
$y = 1$
$z = -3t$

⇒ Of a point where $\kappa \neq 0$, the principal unit normal vector for a smooth curve is

$$\vec{N} = \frac{1}{\kappa} \cdot \frac{d\vec{T}}{ds}$$

if $r(t)$ is a smooth curve, then the principal unit normal vector is,

$$\vec{N} = \frac{1}{|dT/dt|} \times \frac{d\vec{T}}{dt}$$

$$T = \frac{\mathbf{v}}{|v|}$$

Q: Find unit tangent vector and principal unit normal vector of the curve, $r(t) = \cos 2t\,\hat{i} + \sin 2t\,\hat{j}$

$|V(t)| = \sqrt{(-2\sin 2t)^2 + (2\cos 2t)}$

$$T = \frac{\bar{v}}{|v|} = -\sin 2t\,\hat{i} + 2\cos 2t\,\hat{j}$$

$$\frac{dT}{dt} = -2\cos 2t\,\hat{i} - 2\sin 2t\,\hat{j}$$

$$|dT/dt| = \sqrt{(-2\cos 2t)^2 + (-2\sin 2t)^2} = \sqrt{4} = \underline{2}$$

$$\bar{N} \cdot \bar{T} = -\cos 2t \times -\sin 2t + -\sin 2t \times \cos 2t$$

$= \underline{0}$

to prove it is $\perp^r$.

Q: Find the curvature, unit tangent vector and principal normal unit tangent vector of the curve,

$r(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\hat{k}$ * ellipse

$V(t) = -\sin t\,\hat{i} + \cos t\,\hat{j} + 1\hat{k}$

$|V| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1} = \sqrt{2}$

$$T = \frac{\bar{v}}{|v|} = \frac{-\sin t\,\hat{i} + \cos t\,\hat{j} + 1\hat{k}}{\sqrt{2}} = -\frac{\sin t}{\sqrt{2}}\hat{i} + \frac{\cos t}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$$

$$\frac{dT}{dt} = \frac{d}{dt}\left(\frac{\sin t}{\sqrt{2}}\hat{i} + \frac{\cos t}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}\right)$$

$$= -\frac{\cos t}{\sqrt{2}}\hat{i} - \frac{\sin t}{\sqrt{2}}\hat{j} + 0\hat{k}$$

$$= -\frac{\cos t}{\sqrt{2}}\hat{i} - \frac{\sin t}{\sqrt{2}}\hat{j}$$

$$K = \frac{1}{|V|}\left|\frac{dT}{dt}\right|$$

$$= \frac{1}{\sqrt{2}} \times \sqrt{\left(\frac{\cos t}{\sqrt{2}}\right)^2 + \left(\frac{-\sin t}{\sqrt{2}}\right)^2} \qquad \frac{1}{2} + \frac{1}{2}$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2}$$

Since τ is not given.

$$\vec{N} = \frac{1}{\left|\frac{dT}{dt}\right|} \cdot \frac{dT}{dt}$$

$$= \frac{1}{1/\sqrt{2}} \times -\frac{\cos t}{\sqrt{2}}\hat{i} - \frac{\sin t}{\sqrt{2}}\hat{j}$$

$$= \sqrt{2} \cdot -\cos t\,\hat{i} - \sin t\,\hat{j} \;//$$

Q. Find the ~~curvature~~ K, N & T

$$r(t) = (\cos t + t\sin t)\hat{i} + (\sin t - t\cos t)\hat{j} + 3\hat{k}$$

$$V(t) = (-\sin t + t\cos t)\hat{i} + (\cos t + t\sin t)\hat{j} + 0\hat{k}$$

$$|V| = \sqrt{(-\sin t + t\cos t)^2 + (\cos t + t\sin t)^2}$$

$$= \sqrt{\sin^2 t + t^2\cos^2 t + \cos^2 t + t^2\sin^2 t}$$

$$= \sqrt{1 + t^2}$$

$$V(t) = (\cos t + t\cos t + \sin t)\hat{i} + (\cos t - t\sin t + \cos t)\hat{j}$$

$$= (-\sin t + t\cos t + \sin t)\hat{i} + \cos t + (t\sin t + \cos t)\hat{j}$$

$$= t\cos t\,\hat{i} + (\cos t - t\sin t - \cos t)\hat{j}$$

$$V(t) = t\cos t\,\hat{i} + t\sin t\,\hat{j}$$

$$|V| = \sqrt{t^2(\cos^2 t + \sin^2 t)} = t$$

$$T = \frac{\vec{V}}{|V|} = \frac{t\cos t\,\hat{i} + t\sin t\,\hat{j}}{t} \;// = \cos t\,\hat{i} + \sin t\,\hat{j}$$

$$K = \frac{1}{|V|} \cdot \left|\frac{dT}{dt}\right|$$

$$\frac{dT}{dt} = \frac{t\cos t\,\hat{i}}{t} - \frac{t\sin t\,\hat{j}}{t}$$

$$= \cos t\,\hat{i} - \sin t\,\hat{j} - \sin t\,\hat{i} - \cos t\,\hat{j}$$

$$K = \frac{1}{t}(\cos t\,\hat{i} - \sin t\,\hat{j}) \;//$$

$$K = \frac{1}{t}\left|(-\sin t\,\hat{i} + \cos t\,\hat{j})\right|$$

$$K = \frac{1}{t} \times 1 = \frac{1}{t}$$

$$N = \frac{1}{|(dT/dt)|} \cdot \frac{dT}{dt} = -\sin t\,\hat{i} + \cos t\,\hat{j}$$

Q: Show that curvature of a smooth curve $r(t) = f(t)\hat{i} + g(t)\hat{j}$ for f & g are twice differentiable, is given by $K = \dfrac{|f'g'' - g'f''|}{\left[(f')^2 + (g')^2\right]^{3/2}}$

$$r(t) = f(t)\,\hat{i} + g(t)\,\hat{j}$$

$$V(t) = f'(t)\,\hat{i} + g'(t)\,\hat{j}$$

$$|V| = \sqrt{(f'(t))^2 + (g'(t))^2}$$

$$T = \frac{V}{|V|} = \frac{f'(t)\,\hat{i} + g'(t)\,\hat{j}}{\left((f'(t))^2 + (g'(t))^2\right)^{1/2}}$$

$$\frac{dT}{dt} = \frac{f'}{\left((f')^2 + (g')^2\right)^{1/2}}\,\hat{i} + \frac{g'(t)\cdot}{\left((f')^2 + (g'(t))^2\right)^{1/2}}\,\hat{j}$$

$$\frac{dT}{dt} = \frac{\left((f')^2 + (g')^2\right)^{1/2} \times f'' - f' \times \frac{1}{2\sqrt{(f')^2 + (g')^2}}}{\left(((f')^2 + (g')^2)^{1/2}\right)^2}\,\hat{i}$$

$$+ \frac{\left((f')^2 + (g')^2\right)^{1/2} \times g'' - g' \cdot \frac{1}{2\sqrt{(f')^2 + (g')^2}}}{\left(((f')^2 + (g')^2)^{1/2}\right)^2}\,\hat{j}$$

$$\frac{dT}{dt} =$$

$$= \frac{\left((f')^2 + (g')^2\right)^{1/2} \times f'' - \dfrac{f'}{2\left((f')^2 + (g')^2\right)^{1/2}}}{(f')^2 + (g')^2}\,\hat{i} +$$

$$\frac{\left((f')^2 + (g')^2\right)^{1/2} \times g'' - \dfrac{g'}{2\left((f')^2 + (g')^2\right)^{1/2}}}{(f')^2 + (g')^2}\,\hat{j}$$

$$= \frac{\left((f')^2 + (g')^2\right)^{1/2} \cdot f'' - \dfrac{f'}{2\left((f')^2 + (g')^2\right)^{1/2}}}{(f')^2 + (g')^2}\,\hat{i} + \frac{\left((f')^2 + (g')^2\right)^{1/2} \cdot g'' - \dfrac{g'}{2\left((f')^2 + (g')^2\right)^{1/2}}}{(f')^2 + (g')^2}\,\hat{j}$$

1/2/26
uesday.

Functions of Several Variables.

Suppose D is the set of all n tuples $(x_1, x_2, x_3 \cdots x_n)$ a real value if n is a rule $f: D \to \mathbb{R}$ that assigns a unique single real number, $w = f(x_1, x_2, \cdots, x_n)$

$f: \mathbb{R}^n \to \mathbb{R}$.

* The largest set of all n-tuples in $\mathbb{R}^n$ for which the defining rule generates real numbers is called Domain.

* Range consists of set of all output values of the Domain.

1. Find the domain of the function;

$$f(x,y) = \sqrt{y-x^2} \qquad \mathbb{R}^2 \to \mathbb{R}$$

$$\text{Domain} = \{(x,y) \mid y - x^2 \geq 0 \text{ or } y \geq x^2\}$$

$$\text{Range} = [0, \infty)$$

2. $f(x,y) = \sin xy$.

$$\text{Domain} = \{(x,y) \mid x, y \in \mathbb{R}^2\}$$

$$\text{Range} = [-1, 1]$$

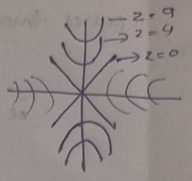
3. $f(x,y,z) = xy \ln(z)$

$$\text{Domain} = \{(x,y,z) \mid z \geq 0\}$$

$$\text{Range} = (-\infty, \infty)$$

~~Interior points~~

$z = y^2 - x^2 \qquad z = 0, 4, 9, -4$



$y^2 - x^2 = 0$ X-axis.

$y^2 - x^2 = 4$ → hyperbola vertical hyperbola

$y^2 - x^2 = 9$ → "

$y^2 - x^2 = -4$ → vertical

Q: Draw the level curves of :

$z = x^2 + y \qquad k = -2, -1, 0, 1, 2$

- $x^2 + y = -2$

  $x^2 = -y - 2$



- $x^2 + y = -1$

  $x^2 = -y - 1$



- $x^2 + y = 0$

  $x^2 = -y$

  $-(x^2) = y$



- $x^2 + y = 1$

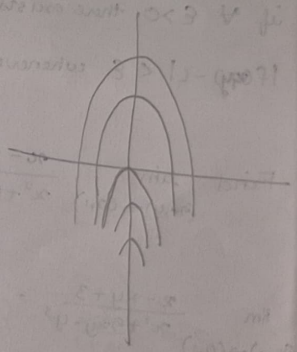
  $x^2 = 1 - y$



- $x^2 + y = 2$

  $x^2 = 2 - y$.





$f(x,y,z) = x^2+y^2-z$. Find an eqn of a level surface passes through point $(1,0,3)$

ie for a value of k, we should get $(1,0,3)$

$\therefore \quad x^2+y^2-z = (1^2+0^2-3)$

$\boxed{x^2+y^2-z=-2}$ ⇒ Eqn

19/02/2026
Thursday.

We say that a function approaches limit 'L' as

$(x,y) \to (x_0,y_0)$;

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L,$$

if $\forall\, \varepsilon > 0$ there exists $\delta > 0$ such that

$|f(x,y) - L| < \varepsilon$ whenever $0 < \sqrt{(x-x_0)^2+(y-y_0)^2} < \delta$

Q:- Find $\lim\limits_{(x,y)\to(0,1)} \dfrac{x-xy+3}{x^2+5xy-y^3}$

$$\lim_{(x,y)\to(0,1)} \frac{x-xy+3}{x^2+5xy-y^3} = \frac{0-0+3}{0^2+0-1} = \underline{\underline{-3}}$$

Q:- A function is said to be continous at (x_0,y_0) if

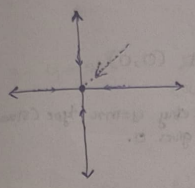
i) $f(x_0,y_0)$ is defined — f^n exists.

ii) $\lim\limits_{(x,y)\to(x_0,y_0)} f(x,y) = L$ — There should be limit

iii) $L = f(x_0,y_0)$

Q:- check the continuity of the function

$$f(x,y) = \begin{cases} \dfrac{2xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$



1.* x-axis ⇒ $(x,0)$

$$\lim_{(x,y)\to(0,0)} \frac{2\times x\times 0}{x^2+0^2} = \underline{\underline{0}}$$

2.* Y-axis ⇒ $(0,y)$

$$\lim_{(x,y)\to(0,0)} \frac{2\times 0\times y}{0^2+y^2} = \underline{\underline{0}}$$

3.* $x=y$

$$\lim_{(x,y)\to(0,0)} \frac{2\times x\times x}{x^2+x^2} = \frac{2x^2}{2x^2} = \underline{\underline{1}}$$

4.* $y = 2x$

$$\lim_{(x,y)\to(0,0)} \frac{2\times x\times 2x}{x^2+4x^2} = \frac{4x^2}{5x^2} = \underline{\underline{\frac{4}{5}}}$$

5.* $y = x^2$

$$\lim_{(x,y)\to(0,0)} \frac{2\times x\times x^2}{x^2+x^4} = \frac{2x^3}{x^2+x^4} = \frac{x^2(2x)}{x^2(1+x^2)} = \underline{\underline{\frac{2x}{1+x^2}}} = \underline{\underline{0}}$$

Two Path Test.

If a function has different limits along two different paths which approaches (x_0, y_0) then,

$$\lim_{(x,y)\to(x_0,y_0)} \text{ does not exist.}$$

Q: Show that the function

$$f(x,y) = \frac{2x^2y}{x^4+y^2} \text{ has no limit at } (0,0).$$

* Any $y=mx$ type (straight line) gives 0.

1. * $y = 2x$

$$\lim_{(x,y)\to(0,0)} \frac{8x^2 \cdot 2x}{x^4+4x^2} = \frac{4x^3}{x^4+4x^2} = \frac{x^2(4x)}{x^2(x^2+4)} = 0$$

2. * $y = x^2$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2 \cdot x^2}{x^4+x^4} = \frac{2x^4}{x^4+x^4} = \frac{x^4(2)}{x^4(2)} = 1$$

$\therefore$ no limit exists.

Q: determine whether limit exists or not.

$$\lim_{(x,y)\to(1,2)} \frac{(x-1)(y-2)}{(x-1)^2+(y-2)^2}$$

$(x,0)$ x

Q Show that $\lim_{(x,y)\to(0,0)} \frac{xy}{|xy|}$ exists or not.

1. $(x,0)$ $\lim_{(x,y)\to(0,0)} \frac{x\times 0}{|x\times 0|} = \frac{0}{0}$

2. $y = 2x$ $\lim_{(x,y)\to(0,0)} \frac{x\times 2x}{|x\times 2x|} = \frac{2x^2}{2x^2} = 1$

3. $y = x^2$ $\lim_{(x,y)\to(0,0)} \frac{x\times x^2}{|x\times x^2|} = \frac{x^3}{x^3} = 1$ if $x>0$; -1 if $x<0$

4. $y = -x$ $\lim_{(x,y)\to(0,0)} \frac{x\times -x}{|x(x-x)|} = \frac{-x^2}{x^2} = -1$

$\therefore$ limit does not exists.

Q Show that the limit does not exists.

$$\lim_{(x,y,z)\to(0,0,0)} \frac{x^2}{x^2+y^2+z^2}$$

1. x axis; $y=0$ $z=0$

$$\lim_{(x,y,z)\to(0,0,0)} \frac{x^2}{x^2+0+0} = 1$$

2. $y = x^2$, $z = 0$

$$\lim_{(x,y,z)\to(0,0,0)} \frac{x^2}{x^2+x^2+0} = \frac{x^2}{2x^2} = \frac{1}{2}$$

$\therefore$ limit doesn't exists.

Q: Check the continuity of the f^n;

$$f(x,y) = \begin{cases} \frac{x^2y}{x^4+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

1. $y = x^2$

$$\lim_{(x,y)\to(0,0)} \frac{x^2 \cdot x^2}{x^4+x^4} = \frac{x^4}{2x^4} = \frac{1}{2}$$

2. $y = 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^2 \cdot 0}{x^4+0} = \frac{x^2}{x^4} = \frac{1}{x^2} = \frac{1}{0}$$

3. $y = x$

$$\lim_{(x,y)\to(0,0)} \frac{x^2 \cdot x}{x^4+x^2} = \frac{x^3}{x^4+x^2} = \frac{x^2 \cdot x}{x^2(x^2+1)} = \frac{x}{x^2+1} = 0$$

4. $y = 2x$

$$\lim_{(x,y)\to(0,0)} \frac{x^2 \cdot 2x}{x^4+4x^2} = \frac{2x^3}{x^4+4x^2} = \frac{2 \cdot x^2 \cdot x}{x^2(x^2+4)} = \frac{2x}{x^2+4}$$

$$= 0$$

24/02/26 Tuesday

The partial derivative of $f(x,y)$ w.r.t x at the point (x_0, y_0) is

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \lim_{h\to 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h}$$

provided limit exists.

* The partial derivative of $f(x,y)$ w.r.t y at the point (x_0, y_0)

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \lim_{h\to 0} \frac{f(x_0, y_0+h) - f(x_0, y_0)}{h}$$

Q: Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ of;

$$f(x,y) = xe^y$$

$\frac{\partial f}{\partial x}$ → keeping y as constant.

$$\frac{\partial f}{\partial x} = 1 \times e^y$$

$$\frac{\partial f}{\partial x} = e^y$$

$\frac{\partial f}{\partial y}$ → keeping x as constant.

$$\frac{\partial f}{\partial y} = x \times e^y$$

$$\frac{\partial f}{\partial y} = xe^y$$

2. Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial z}$ of f^n:

$f(x,y,z) = x^2 \sin(yz)$

$\frac{\partial f}{\partial x}$ ⇒ Keeping y & z const.

when taking w.r.t x, other y & z are all just constants.

* $\frac{\partial f}{\partial x} = 2x^{2-1} \sin(yz)$ //

* $\frac{\partial f}{\partial y} = x^2 \times \cos(yz) \cdot z$ //

* $\frac{\partial f}{\partial z} = x^2 \times \cos(yz) \cdot y$ //

7, 9, 10, 15, 20

23, 26, 30, 31, 32.

HW

7. $f(x,y) = \sqrt{x^2+y^2}$ $\quad (x^2+y^2)^{1/2}$

$\frac{\partial f}{\partial x} = \frac{1}{2} \times (x^2+y^2)^{-1/2} \times 2x$ $\quad \frac{1}{2}(x^2+y^2)^{1/2-1}$

$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2+y^2}}$ //

$\frac{\partial f}{\partial y} = \frac{1}{2}(x^2+y^2)^{-1/2} \times 2y$

$\frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2+y^2}}$

9. $f(x,y) = \frac{1}{(x+y)}$ $\quad x+y$

$\frac{\partial f}{\partial x} = \ln(x+y) \times 1$ $\quad \frac{\partial f}{\partial y} = \ln(x+y)$ //

$\frac{\partial f}{\partial x} = \ln(x+y)$ //

10. $f(x,y) = \frac{x}{(x^2+y^2)}$

$\frac{\partial f}{\partial x} = \frac{(x^2+y^2) \times 1 - x \times 2x}{(x^2+y^2)^2} = \frac{x^2+y^2-2x^2}{x^4+y^4} = \frac{y^2-x^2}{(x^4+y^4)^2}$

$\frac{\partial f}{\partial y} = -(x^2+y^2) \times$

$\frac{\partial f}{\partial y} = \ln(x^2+y^2) \times 2y$

$\frac{\partial f}{\partial y} = \frac{(x^2+y^2) \times 0 - x \times 2y}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$ //

15. $f(x,y) = \ln(x+y)$

$\frac{\partial f}{\partial x} = \frac{1}{x+y} \times 1$ $\quad \frac{\partial f}{\partial y} = \frac{1}{x+y} \times 1$

$\frac{\partial f}{\partial x} = \frac{1}{x+y}$ // $\quad \frac{\partial f}{\partial y} = \frac{1}{x+y}$ //

~~20. $f(x,y) = x\log y^x$~~

~~$\frac{d(\log_a x)}{dx} = \frac{1}{x\ln a}$~~

~~$\frac{\partial f}{\partial x} = \frac{1}{x \ln y^x}$~~

20. $f(x,y) = \log_y x$

$f(x,y) = \frac{\log x}{\log y}$

~~$\frac{\partial f}{\partial y} = \frac{\log x}{y}$~~

~~$f_y = \frac{\log x}{y}$~~

~~$f_y = \frac{\log x}{1/x} = x\log x$~~

~~$f_x = \frac{1/x}{\log y} = \frac{1}{x\log y}$~~

23. $f(x,y,z) = 1 + xy^2 - 2z^2$

$f_x = \frac{\partial f}{\partial x} = 0 + 1 \times y^2 - 2z^2 = y^2 - 2z^2$ // $= y^2$

$f_y = 0 + x \times 2y - 2z^2 = 2xy - 2z^2$ // $2xy$

$f_z = 0 + 0 - 4z = -4z$.

~~24. $f(x,y,z) = (x^2+y^2+z^2)^{-1/2}$~~

~~$f_x = -\frac{1}{2} \times 2x = -x$~~

~~$f_y = -\frac{1}{2} \times (2y)$~~

~~$f_x = -\frac{1}{2} \times (2x)^{-3/2}$~~

~~$= -x^{-3/2}$~~

~~$-\frac{1}{2} - 1$~~

~~$f_y = -\frac{1}{2} \times (2y)^{-3/2}$~~

26. $f(x,y,z) = (x^2+y^2+z^2)^{-1/2}$

* $f_x = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \times 2x$

$= \frac{-x}{(x^2+y^2+z^2)^{3/2}}$ //

* $f_y = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \times 2y$

$= \frac{-y}{(x^2+y^2+z^2)^{3/2}}$ //

* $f_z = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \times 2z$

$= \frac{-z}{(x^2+y^2+z^2)^{3/2}}$ //

30. $f(x,y,z) = yz\ln(xy)$

* $f_x = yz \times \frac{1}{xy} \times y \times 1 = \frac{zy}{x}$ //

* $f_y = yz \times \frac{1}{xy} \times x \times 1 + \ln(xy) \times z$

$= z + z\ln(xy)$ //

~~$f_y = -\frac{1}{2} \times (2y)^{-3/2}$~~

~~$\frac{d}{dy}\left(\frac{\log x}{\log y}\right)$~~

~~$\frac{\log x}{\log y}$~~ ~~$\frac{1}{\log y}$~~

~~$\log x$~~ ~~$\frac{1}{1/y}$~~

~~$y \log x$~~

~~$\frac{\log x}{\log y}$~~

~~$\log x \cdot \log y$~~

* $f_z = y \ln(xy) \times 1 = y\ln(xy)$

31. $f(x,y,z) = e^{-(x^2+y^2+z^2)}$

$f_x = -2x \times e^{-(x^2+y^2+z^2)}$

$f_y = -2y \times e^{-(x^2+y^2+z^2)}$

$f_z = -2z \times e^{-(x^2+y^2+z^2)}$

32. $f(x,y,z) = e^{-xyz}$

$f_x = -yz \times e^{-xyz}$ //

$f_y = -xz \times e^{-xyz}$ //

$f_z = -xy\, e^{-xyz}$ //

80. $f(x,y) = \log_y x = \frac{\log x}{\log y}$ $\log_y x = \frac{\log x}{\log y}$

$\frac{\partial f}{\partial x} = \frac{1}{x \log y}$

$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(\log x (\log y)^{-1}) = \log x \times -1 \times (\log y)^{-2} \times \frac{1}{y}$

$= \frac{-\log x}{y(\log y)^2}$ //

25/02/20
Wednesday.

* $f(x,y) = \begin{cases} 0 & xy \neq 0 \\ 1 & xy = 0 \end{cases}$

$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x,y)}{h}$

$= \frac{f(h,0) - f(0,0)}{h} = \frac{1-1}{h} = 0$

$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x,y)}{h}$

$= \frac{f(0,h) - f(0,0)}{h} = \frac{1-1}{h} = 0$

continuity.

Second Partial Derivative

Suppose f is a fⁿ of two variables x & y, then the partial derivatives will also be the functions of x & y. So they themselves will have partial derivatives.

Q: Find // $f(x,y) = x\cos y + ye^x$

$\frac{\partial f}{\partial x} = \cos y + ye^x$

$\frac{\partial f}{\partial y} = -x \sin y + e^x$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$

$$= \frac{\partial}{\partial x}(\cos y + y e^x)$$

$$= 0 + y e^x$$

$$= \underline{y e^x}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}(-x \sin y + e^x)$$

$$= 0 - x \cos y + 0$$

$$= \underline{-x \cos y}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

$$= \frac{\partial}{\partial x}(-x \sin y + e^x)$$

$$= \underline{-\sin y + e^x}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

$$= \frac{\partial}{\partial y}(\cos y + y e^x)$$

$$= \underline{-\sin y + e^x}$$

Q: Find all second order partial derivative of the f^n.

$$f(x,y) = x \ln(xy)$$

$$\frac{\partial f}{\partial x} = x \times \frac{1}{xy} \times y + \ln(xy) = \underline{\frac{y}{y} + \ln(xy)} = \underline{1 + \ln(xy)}$$

$$\frac{\partial f}{\partial y} = x \times \frac{1}{xy} \times x = \underline{\frac{1}{y}} \quad \underline{\frac{x}{y}}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$

$$= \frac{\partial}{\partial x}\left(1 + \ln(xy)\right)$$

$$= 0 + \frac{1}{xy} \times y$$

$$= \underline{\frac{1}{x}}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

$$= \frac{\partial}{\partial y}\left(\frac{x}{y}\right)$$

$$= \frac{\partial}{\partial y}(xy^{-1})$$

$$= -1 \times (xy)^{-2} \times x$$

$$= x \times -1 \times y^{-2} \times 1$$

$$= -x/y^2$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

$$= \frac{\partial}{\partial x}\left(x/y\right)$$

$$= \underline{\underline{\frac{1}{y}}}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

$$= \frac{\partial}{\partial y}\left(1 + \ln(xy)\right)$$

$$= 0 + \frac{1}{xy} \times x$$

$$= \underline{\underline{\frac{1}{y}}}$$

* If f and its partial derivatives $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial^2 f}{\partial x \partial y}$, $\frac{\partial^2 f}{\partial y \partial x}$ are defined, and continous around a point (x_0, y_0) then, $\left.\frac{\partial^2 f}{\partial x \partial y}\right|_{(x_0,y_0)} = \left.\frac{\partial^2 f}{\partial y \partial x}\right|_{(x_0,y_0)}$

Q: Find $\frac{\partial^2 w}{\partial x \partial y}$ of; $w = xy + \frac{e^y}{y^2+1}$

$$\frac{\partial w}{\partial y} = x + \frac{(y^2+1)\cdot e^y - e^y(2y)}{(y^2+1)^2} = \frac{x + y^2 e^y + e^y - 2ye^y}{(y^2+1)^2}$$

$$\frac{\partial w}{\partial x} = y$$

$$\frac{\partial}{\partial y}\left(\frac{\partial w}{\partial x}\right) = 1$$

$$\frac{\partial}{\partial x}\left(\frac{\partial w}{\partial y}\right) = 1$$

∴ By the theorem.

Q: Find all second ordered partial derivatives of ;

1. $g(x,y) = x^2y + \cos y + y \sin x$
2. $g(x,y) = \ln(x+y)$
3. $w = \frac{x-y}{x^2+y^2}$

1. $g(x,y) = x^2y + \cos y + y\sin x$

$$\frac{\partial g}{\partial x} = 2xy + 0 + y\cos x = 2xy + y\cos x$$

$$\frac{\partial g}{\partial y} = x^2 - \sin y + \sin x$$

$$\frac{\partial^2 g}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial g}{\partial x}\right) = \frac{\partial}{\partial x}(2xy + y\cos x) = 2y - y\sin x$$

$$\frac{\partial^2 g}{\partial y^2} = \frac{\partial}{\partial y}\left(\partial g/\partial y\right) = \frac{\partial}{\partial y}(x^2 - \sin y + \sin x) = -\cos y$$

$$\frac{\partial^2 g}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\partial g/\partial x\right) = \frac{\partial}{\partial y}(2xy + y\cos x) = 2x + \cos x$$

$\frac{\partial^2 g}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial g}{\partial y}\right)$

2. $w = \ln(x+y)$

$\frac{\partial w}{\partial x} = \frac{1}{x+y}$

$\frac{\partial w}{\partial y} = \frac{1}{x+y}$

$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial w}{\partial x}\right) = -\frac{1}{x^2}$

$\frac{\partial^2 w}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial w}{\partial y}\right) = -\frac{1}{y^2}$

$\frac{\partial w}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial w}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{1}{x+y}\right) = -\frac{1}{x^2}$

x^{-1-2}

$-x^{-2}$

$-\frac{1}{x^2}$

$x^4 + 2x^2y + y^2$

3. $w = \frac{x-y}{x^2+y}$

$\frac{\partial w}{\partial x} = \frac{(x^2+y)\times 1 - (x-y)\times 2x}{(x^2+y)^2}$

$= \frac{x^2+y-2x^2+2xy}{x^4+2x^2y+y^2}$

$= \frac{+2xy - x^2 + y}{x^4+2x^2y+y^2}$

$\frac{\partial w}{\partial y} = \frac{(x^2+y)\times -1 - (x-y)(2x^2+2xy)(-1)}{(x^2+y)^2}$

$= \frac{-x^2-y+x+y}{(x^2+y)^2}$

$= \frac{-x^2+x}{(x^2+y)^2}$

$= \frac{-x-x^2}{(x^2+y)^2}$

4. $w = \frac{x-y}{x^2+y}$

$\frac{\partial w}{\partial x} = \frac{(x^2+y)\times(1) - (x-y)\times(2x)}{(x^2+y)^2} = \frac{x^2+y-2x^2+2xy}{(x^2+y)^2} = \frac{2xy-x^2+y}{(x^2+y)^2}$

$\frac{\partial w}{\partial y} = \frac{(x^2+y)(-1) - (x-y)\times 1}{(x^2+y)^2} = \frac{-x^2-y+x+y}{(x^2+y)^2} = \frac{x-x^2}{(x^2+y)^2}$

$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial w}{\partial x}\right) = \frac{(x^2+y)^2(2-2x) - (2xy-x^2+y)(4x^3+4x)}{(x^2+y)^2}$

$$\star \quad \frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{2xy - x^2 - y}{x^4 + 2x^2y + y^2}\right)$$

$$= \frac{(x^2+y)^2(2x - 0 - 1) - (2xy - x^2 - y)(2x^2 + 2y)}{\left((x^2+y)^2\right)^2}$$

$$= \frac{(x^4 + 2x^2y + y^2)2x - \left[4x^3y - 2x^4 - 2x^2y + 2xy^2 - 2x^2y - 2y^2\right]}{(x^2+y)^4}$$

$$= \frac{2x^5 + 4x^3y + 2xy^2 - 4x^3y + 2x^4 + 2x^2y - 2xy^2 + 2x^2y + 2y^2}{(x^2+y)^4}$$

$$= \frac{2x^5 + 2x^4 + 4x^2y + 2y^2}{(x^2+y)^4}$$ //

$$\frac{\partial^2 w}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{x^2 - x^2}{(x^2+y)^2}\right)$$

$$= \frac{(x^2+y)^2(0) - (x - x^2)(0 + 2x^2 + 2y)}{(x^2+y)^4}$$

$$= \frac{-(x - x^2)(2x^2 + 2y)}{(x^2+y)^4} = \frac{-\left[2x^3 - 2x^4 + 2xy - 2x^2y\right]}{(x^2+y)^4}$$

$$= \frac{2x^4 + 2x^2 + 2x^2y - 2xy - 2x^3 - 2y}{(x^2+y)^4}$$

$$= \frac{-2x^3}{}$$

$$\star \quad \frac{\partial^2 w}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{-x - x^2}{(x^2+y)^2}\right)$$

$$= \frac{(x^2+y)^2(0) - \left[(-x - x^2) \times 2(x^2+y)\right]}{(x^2+y)^4}$$

$$= \frac{-\left[(-x - x^2) - 2x^2 + 2y\right]}{(x^2+y)^4} = \frac{(x^2 + x^2) \times 2(x^2+y)}{(x^2+y)^{\cancel{4}\,3}}$$

$$= \frac{x^2 + 2x^2 + 2x^2 - 2y}{(x^2+y)^4} = \frac{2(x^2+x)}{(x^2+y)^3}$$

$$\star \quad \frac{\partial^2 w}{\partial y \partial x} = \frac{\partial}{\partial x}\left(\frac{\partial w}{\partial y}\right)$$

$$= \frac{(x^2+y)^2 \times (-1 - 2x) - \left[(-x - x^2)(2(x^2+y)\right.}{(x^2+y)^4}$$

$$= \frac{(x^2+y)^2 \times (-1 - 2x) + 4x(x^2+x)}{(x^2+y)^3}$$

$$= \frac{2x^3 - 2xy + 3x^2 - y}{(x^2+y)^3}$$ //

Q:- Find f_{yxyz} F_{yxyz}

$$F = 1 - 2xy^2z + x^2y$$

$$f_{yxyz} = \frac{\partial}{\partial z}\left[\frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]\right]\right] = \frac{\partial^4 f}{\partial z\,\partial y\,\partial x\,\partial y}$$

* $\frac{\partial f}{\partial y} = -2x \times 2y \times z + x^2 = -4xyz + x^2$

* $\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = +\,-4yz + 2x$

* $\frac{\partial^3 f}{\partial y\,\partial x\,\partial y} = -4z + 0 = -4z$

* $\frac{\partial^4 f}{\partial z\,\partial x\,\partial y\,\partial z} = -4$

$\alpha, \beta \rightarrow$ constants

Q:- Show that $u(x,t) = \sin(\alpha x)\cdot e^{-\beta x}$ satisfies one dimentional heat eqn.

Heat eqn $\rightarrow f_t = f_{xx}$

LHS $f_t = 0$

RHS; $f_x = \sin(\alpha x)\cdot -\beta\cdot e^{-\beta x} + e^{-\beta x}\cdot\cos(\alpha x)\times\alpha$

$= -\beta e^{-\beta x}\cdot\sin(\alpha x) + \alpha e^{-\beta x}\cdot\cos(\alpha x)$

$f_{xx} = \{-\beta\cdot e^{-\beta x}\cdot\cos(\alpha x)\cdot\alpha + \sin(\alpha x)\cdot -\beta^2 e^{-\beta x}\} +$

$\alpha e^{-\beta x}\cdot -\sin(\alpha x)\cdot + \cos(\alpha x)\times -\alpha\beta e^{-\beta x}$

$= \left[\beta\alpha e^{-\beta x}\cdot\cos(\alpha x) + \beta^2 e^{-\beta x}\sin(\alpha x)\right] + \left[-\alpha^2 e^{-\beta x}\sin(\alpha x)\right.$

$\left. -\alpha\beta e^{-\beta x}\cos(\alpha x)\right.$

$= \beta^2 e^{-\beta x}\sin(\alpha x) - \alpha^2 e^{-\beta x}\sin(\alpha x) - 2\alpha\beta e^{-\beta x}\cos(\alpha x)$

$= e^{-\beta x}\left[(\beta^2-\alpha^2)\sin(\alpha x) - 2\alpha\beta\cos(\alpha x)\right]$

$\therefore \quad f_t = f_{xx} = 0$

$\therefore \quad e^{-\beta x}\left[(\beta^2-\alpha^2)\sin(\alpha x) - 2\alpha\beta\cos(\alpha x)\right] = 0$

$(\beta^2-\alpha^2)\sin(\alpha x) = 2\alpha\beta\cos(\alpha x)$

$(\beta^2-\alpha^2)\sin(\alpha x) - 2\alpha\beta\cos(\alpha x) = 0$

$\therefore \quad \beta^2-\alpha^2 = 0 \quad \& \quad 2\alpha\beta = 0$

$\beta^2 = \alpha^2$

$\pm\beta = \pm\alpha$

Chain rule :-

Q:- using chain rule ; find the derivative of

$w = xy$ w.r.t t where $x = \cos t$ & $y = \sin t$.

$w = xy$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

* $\frac{dw}{dt} = -y$
* $\frac{\partial w}{\partial x} = y$
* $\frac{\partial w}{\partial y} = x$
* $\frac{dx}{dt} = -\sin t$
* $\frac{dy}{dt} = \cos t$

$$\frac{dw}{dt} = y(-\sin t) + x \cdot \cos t$$

$$= -y \sin t + x \cdot \cos t$$

$$= -(\sin t)^2 + (\cos t)^2$$

$$= -\sin^2 t + \cos^2 t$$

Q: $w = x^2 + y^2$

$x = \cos t + \sin t \qquad y = \cos t - \sin t \qquad t = 0$

* $\frac{\partial w}{\partial x} = 2x$
* $\frac{dx}{dt} = -\sin t + \cos t$
* $\frac{\partial w}{\partial y} = 2y$
* $\frac{dy}{dt} = -\sin t - \cos t$

$$\frac{dw}{dt} = 2x \times (-\sin t + \cos t) + 2y(-\sin t - \cos t)$$

$$= 2x(\cos t + \sin t)(-\sin t + \cos t) + 2(\cos t - \sin t)(-\sin t -$$

$$= 2x(\cos^2 t - \sin^2 t) + 2(\cos t - \sin t)(-\cos t - \sin t)$$

$$= 2x(\cos^2 t - \sin^2 t) + 2(\cos^2 t - \sin^2 t)$$

$$= 0$$

Q: $w = xy + z \qquad x = \cos t \qquad y = \sin t \qquad z = t$

$\frac{\partial w}{\partial x} = y \qquad \frac{dx}{dt} = -\sin t$

$\frac{\partial w}{\partial y} = x \qquad \frac{dy}{dt} = +\cos t$

$\frac{\partial w}{\partial z} = 1 \qquad \frac{dz}{dt} = 1$

$$\frac{dw}{dt} = y \cdot \sin t + x \times -\cos t + 1 \times 1$$

$$= -\sin^2 t + + \cos^2 t + 1$$

$$= \cos^2 t - \sin^2 t + 1$$

$$= \underline{\underline{2}}$$

$\sin^2 t + \cos^2 t = 1$

$\sin^2 t = 1 - \cos^2 t$

$\cos^2 t - 1 + \cos^2 t$

Q: Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$

where $w = x^2 + y^2$ $\quad x = r - s$ $\quad y = r + s$.

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$* \frac{\partial w}{\partial x} = 2x$ $\quad * \frac{\partial x}{\partial r} = 1$

$* \frac{\partial w}{\partial y} = 2y$ $\quad * \frac{\partial y}{\partial r} = 1$

$$\frac{\partial w}{\partial r} = 2x \times 1 + 2y \times 1 = 2(r-s) + 2(r+s)$$

$$= 2r - 2s + 2r + 2s$$

$$= 4r //$$

$* \frac{\partial x}{\partial s} = -1$

$* \frac{\partial y}{\partial s} = 1$

$$\frac{\partial w}{\partial s} = 2x \times -1 + 2y \times 1$$

$$= 2(r-s) + 2(r+s)$$

$$= -2r + 2s + 2r + 2s$$

$$= \underline{\underline{4s}}$$

Q: Find $\frac{dy}{dx}$ if $y^2 - x^2 - \sin xy = 0$

$$f(x,y) = y^2 - x^2 - \sin xy = 0$$

$$\frac{df}{dx} = -2x - \cos(xy) \times y$$

$$\frac{\partial f}{\partial y} = +2y - \sin(xy) \times x$$

$$\frac{dy}{dx} = \frac{-f_x}{-f_y}$$

$$\frac{dy}{dx} = \frac{-(-2x - y\cos(xy))}{2y - x\sin(xy)}$$

$$= \frac{2x + y\cos(xy)}{2y - x\sin(xy)} //$$

3/26
Friday.

Find the directional derivative of the f^n;

$f(x,y) = x^2 + xy$ at $(1,2)$

$\bar{u} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

$$\left(\frac{df}{ds}\right)_{u,P_0} = \lim_{s\to 0} \frac{f(x_0+su_1, y_0+su_2) - f(x_0,y_0)}{s}$$

$x(s) = 1 + \frac{s}{\sqrt{2}}$

$y(s) = 2 + \frac{s}{\sqrt{2}}$

$$\lim_{s\to 0} \frac{f(1+s/\sqrt{2},\ 2+s/\sqrt{2}) - f(1,2)}{s}$$

$$= \lim_{s\to 0} \frac{(1+s/\sqrt{2})^2 + (1+s/\sqrt{2})(2+s/\sqrt{2}) - 3}{s}$$

$$= \frac{(1 + \frac{s^2}{2} + \sqrt{2}s) + (2 + \frac{s}{\sqrt{2}} + \sqrt{2}s + \frac{s^2}{2}) - 3}{s}$$

$$= \frac{s^2 + 2\sqrt{2}s + s/\sqrt{2}}{s}$$

$$= \frac{s(s + 2\sqrt{2} + 1/\sqrt{2})}{s} \quad = 0 + 2\sqrt{2} + 1/\sqrt{2}$$

$$= \frac{4+1}{\sqrt{2}} = 5/\sqrt{2}$$

$$\star \frac{df}{ds} = \frac{df}{dx}\cdot\frac{dx}{ds} + \frac{df}{dy}\cdot\frac{dy}{ds}$$

$x = x_0 + su_1$

$y = y_0 + su_2$

$$= \frac{\partial f}{\partial x}\cdot u_1 + \frac{\partial f}{\partial y}\cdot u_2$$

$$= \left(\frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}\right)\cdot(u_1\hat{i} + u_2\hat{j})$$

$\nabla f \rightarrow$ gradient vector.

So for the ans, ...

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}$$

$$= (2x+y)\hat{i} + x\hat{j}$$

$$= (2\times 1 + 2)\hat{i} + 1\hat{j}$$

$$= 4\hat{i} + 1\hat{j}$$

$u = 1/\sqrt{2}\,\hat{i} + 1/\sqrt{2}\,\hat{j}$

$$\nabla f \cdot u = \frac{4}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{5}{\sqrt{2}}$$

Q: Find the derivative of the function.

$f(x,y) = x\cdot e^y + \cos(xy)$ at the point $(2,0)$ along the direction $v = 3\hat{i} - 4\hat{j}$.

change v into unit normal vector;

$|\vec{v}| = \sqrt{3^2+4^2} = \sqrt{25} = 5$

$\therefore u = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}$$

$$= \left[(x \cdot e^y + e^y \times 1) + -\sin(xy)\right]\hat{i} + \left[(x \cdot e^y + e^y) + -\sin(xy)\right]\hat{j}$$

$$= \left[e^y(x - \sin(xy))\right]\hat{i} + \left[e^y(x - \sin(xy))\right]\hat{j}$$

$$= e^y\hat{i} + e^y\hat{j}$$

$$= e^y\hat{i} + e^y\hat{j}$$

$$= 1\hat{i} + \hat{j}$$

$$u = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

$$\nabla f \cdot u = (1\hat{i} + \hat{j}) \cdot (3/5\,\hat{i} + 4/5\,\hat{j})$$

$$= \frac{3}{5} + \frac{4}{5}$$

$$= 7/5$$

Q:- Find the directions in which $f(x,y) = \frac{x^2}{2} + \frac{y^2}{2}$

i] increases most rapidly at the point $(1,1)$,

ii] decreases "

iii] What are the direction of zero change at $(1,1)$.

$$f(x,y) = \frac{x^2}{2} + \frac{y^2}{2}$$

$$\nabla f|_{(1,1)} = x\hat{i} + y\hat{j}$$

$$= 1\hat{i} + \hat{j}$$

$$D_u f = \sqrt{1^2 + 1^2} = \sqrt{2}$$

i) unit vector;

$$\vec{u_1} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

ii) $u_2 = -\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$

iii) $\vec{u_3} = a\hat{i} + b\hat{j}$

$u_1 \cdot u_3 = 0 \qquad \sqrt{a^2 + b^2} = 1$

$$= \frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} \quad \text{or} \quad -\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

$$\boxed{D_f(u) = \nabla f \cdot u}$$

Q:- $f(x,y) = \frac{(x-y)}{x+y}$. Find

$(-\frac{1}{2}, \frac{3}{2})$ for which

i] $D_u f(-\frac{1}{2}, \frac{3}{2})$ is largest.

ii] " smallest.

iii] 0 " zero.

iv] " $= -2$

v] & $D_u f = 1$

ans: $\nabla f = \left(\frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}\right)$ $\quad f = \frac{(x-y)}{(x+y)}$

$$\frac{\partial f}{\partial x} = \frac{(x+y)(1-0) - (x-y)(1+0)}{(x+y)^2}$$

$$\frac{\partial f}{\partial x} = \frac{(x+y) - (x-y)}{(x+y)^2} = \frac{x+y-x+y}{(x+y)^2} = \frac{2y}{(x+y)^2}$$

$$\frac{\partial f}{\partial y} = \frac{(x+y)\times(-1) - (x-y)(1)}{(x+y)^2} = \frac{(-x-y)-(x-y)}{(x+y)^2}$$

$$= \frac{-x-y-x+y}{(x+y)^2}$$

$$= \frac{-2x}{(x+y)^2}$$

$$\nabla f = \frac{2y}{(x+y)^2}\hat{i} - \frac{2x}{(x+y)^2}\hat{j}$$

$$\nabla f\Big|_{(-\frac{1}{2}, \frac{3}{2})} = \frac{2\times\frac{3}{2}}{(-\frac{1}{2}+\frac{3}{2})^2}\hat{i} - \frac{(2\times-\frac{1}{2})}{(-\frac{1}{2}+\frac{3}{2})^2}\hat{j}$$

$$= 3\hat{i} + 1\hat{j}$$

$$D_u f = \sqrt{3^2+1^2} = \sqrt{9+1} = \sqrt{10}$$

i] $u_1 = \frac{3}{\sqrt{10}}\hat{i} + \frac{1}{\sqrt{10}}\hat{j}$

ii] Smallest $\Rightarrow$ $u_2 = \frac{-3}{\sqrt{10}}\hat{i} + \frac{1}{\sqrt{10}}\hat{j}$

iii] $u_3 = a\hat{i} + b\hat{j}$

$u_1 \cdot u_3 = 0$ $\quad \sqrt{a^2+b^2} = 1$

$u_3 = \frac{3}{\sqrt{10}}\hat{i} - \frac{1}{\sqrt{10}}\hat{j}$ or $-\frac{3}{\sqrt{10}}\hat{i} + \frac{1}{\sqrt{10}}\hat{j}$

iv) $\nabla f \cdot (u_4\hat{i} + u_4\hat{j}) = -2$ $\quad \sqrt{u_4^2 + u_4^2} = 1$

$(3\hat{i}+1\hat{j}) \cdot (a\hat{i} + b\hat{j}) = -2$ $\quad \sqrt{a^2+b^2} = 1$

$a^2+b^2 = 1$

$3a + b = -2$ $\quad b^2 = (1-a^2)$

Q: Is there any direction u in which rate of change of $f(x,y)$ is equal to $x^2 - 3xy + 4y^2$ at $(1,2)$ equals 14.

$f = x^2 - 3xy + 4y^2$

∇f

$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}$

$\nabla f = (2x - 3y)\hat{i} + (-3x + 8y)\hat{j}$

$\nabla f|_{(1,2)} = (2-6)\hat{i} + (-3+16)\hat{j}$

$\nabla f|_{(1,2)} = -4\hat{i} + 13\hat{j}$

$16+100$
$\sqrt{160}$

$D_{\vec{u}}f = \sqrt{(-4)^2 + 10^2} = \sqrt{185}$

$169, 16$
185

$\vec{u} = \frac{-4}{\sqrt{185}}\hat{i} + \frac{10}{\sqrt{185}}\hat{j}$ max

$\vec{u} = \frac{4}{\sqrt{160}}\hat{i} \quad \frac{10}{\sqrt{160}}\hat{j}$ min

$\vec{u} = \frac{4}{\sqrt{185}}\hat{i} - \frac{10}{\sqrt{185}}\hat{j}$

so the largest unit vec = $\frac{}{\sqrt{}}$

So the largest vector is $-4\hat{i} + 13\hat{j}$ with mag =

" Smallest vector is $4\hat{i} - 13\hat{j}$ with mag = 13

So no vector with mag 14 is possible.

iv) $\nabla f \cdot (u_1, u_2) = -2$

$(3,1)(u_1, u_2) = -2$

$3u_1 + u_2 = -2$

$u_2 = -2 - 3u_1$

$u_1^2 + u_2^2 = 1$

$u_1^2 + (-2 - 3u_1)^2 = 1$

$u_1^2 + (-2)^2 + (-2 \times 3 \times 2) + \ldots$

$u_1^2 + 4 + 12u_1 + 9u_1^2 = 1$

$10u_1^2 + 12u_1 + 3 = 1$

$10u_1^2 + 16 = 1$

$10u_1^2 + 16 - 1 = 0$

$10u_1^2 + 15 = 0$

$10u_1^2 = -15$

$u_1^2 = -1.5$

$u_1 = \frac{-12 \pm \sqrt{144 - 120}}{20}$

$= \frac{-12 \pm \sqrt{24}}{20}$

$= \frac{-6 \pm \sqrt{6}}{10}$

$= -0.85$ or $= -0.05$

$u_2 = -2 - 3u_1$ or $u_2 = -4.52$

$= -0.95$

Q: Find rate of change of f^n.

$f(x,y) = x^2+y^2$ along the curve, $r(t) = t\hat{i} + t^2\hat{j}$

$$f(x,y) = x^2 + y^2$$

$$\nabla f = 2x\,\hat{i} + 2y\,\hat{j}$$

$$\frac{dr}{dt} = v(t) = 1\hat{i} + 2t\hat{j}$$

$$\frac{dw}{dt} = \vec{\nabla f} \cdot \vec{v}$$

$$= 2x + 4yt$$

$$x \to t$$
$$y \to t^2$$

$$\frac{dw}{dt} = 2(t) + 4(t^2)\,t$$

$$= 2t + 4t^3$$

Q: Suppose that the temperature distribution of a 3D object is $T(x,y,z) = xyz$. Compute the rate of change along

$$r(t) = t\hat{i} + (1+t)\hat{j} + t^2\hat{k}$$

$$f(x,y,z) = x \cdot y \cdot z$$

$$\nabla f = \frac{df}{dx}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$\nabla f = yz\,\hat{i} + xz\,\hat{j} + xy\,\hat{k}$$

$$r(t) = t\hat{i} + (1+t)\hat{j} + 2t^2\hat{k}$$

$$v(t) = 1\hat{i} + 1\hat{j} + 2t\hat{k}$$

$$\frac{dw}{dt} = \nabla f \cdot \vec{v}$$

$$= yz + xz + xy \cdot 2t$$

$x \to t$

$y \to 1+t$

$z \Rightarrow t^2$

$$= (1+t)t^2 + t \times t^2 + t \times t^2 \times 2t$$

$$= t^2 + t^3 + t^3 + 2t^4$$

$$= 2t^4 + 2t^3 + t^2$$